

AI Translation and Image-Analysis Platform

Subscription-gated text, speech, and image AI tools in one full-stack application

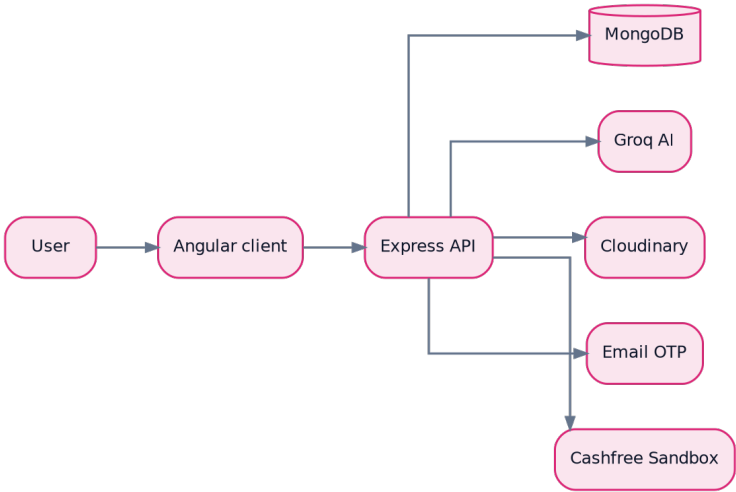
Project context: Suhane Safar internal technology exploration project	Status: Demo application - sandbox payments
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A full-stack demonstration that verifies accounts with email OTP, offers Cashfree Sandbox subscription plans, and provides AI features for text translation, voice translation and playback, and image interpretation through a single account.

Technology stack

Angular	Node.js / Express	MongoDB	Groq AI APIs
Cloudinary	Cashfree Sandbox	JWT	Email OTP

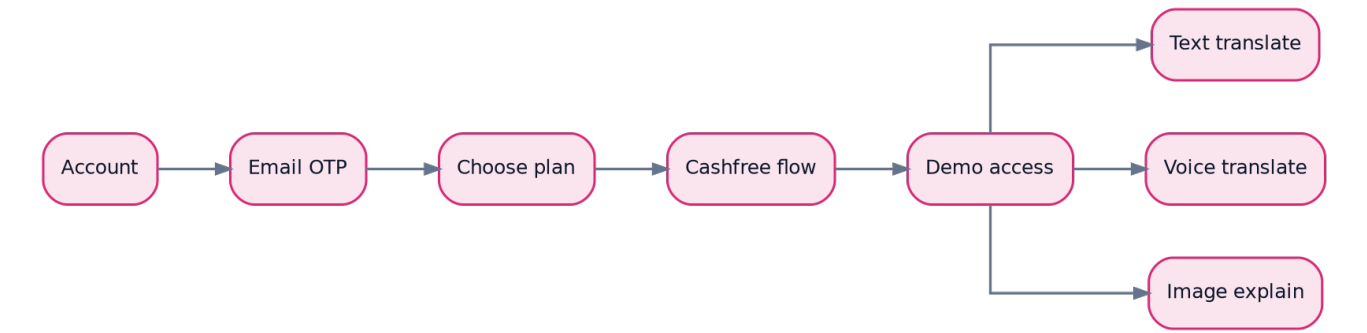
System at a glance



Text workflow Select languages, translate written text, and generate speech output.	Audio workflow Record speech, transcribe or translate it, and play the result.
Image workflow Upload an image and receive an AI-generated explanation.	Subscription journey Connects account verification, plan selection, payment flow, and access state.

System design and component ownership

Angular owns the browser experience while Express coordinates the database and external services. MongoDB stores user and subscription data; Groq handles AI operations; Cloudinary supports image handling; Cashfree remains a sandbox payment provider.



Component	Responsibility
Angular client	Account, plans, translation, audio, and image screens.
Express API	Coordinates authentication, subscriptions, uploads, and AI requests.
MongoDB	Stores user identity, verification, and subscription state.
Groq AI	Performs language, speech, and image-understanding operations.
Cloudinary	Provides temporary hosted media for image processing.
Cashfree + email	Demonstrates subscription authorization and OTP delivery.

Key design decisions

- Keep the application explicitly labeled as a sandbox demonstration.
- Use one authenticated account to expose multiple AI input types.
- Separate frontend interactions from third-party API coordination in Express.
- Document current payment and authorization gaps rather than presenting them as production-ready.

Core execution flow

A new user verifies an email address, selects a plan, completes the sandbox payment flow, and then invokes one of three AI paths. Each path converts browser input into an external-service request and presents the normalized result.



Implementation notes

- The user journey links signup, email OTP, plan selection, payment, and AI access.
- Text, audio, and image inputs have separate interface flows but share account state.
- The backend stores users and subscription records in MongoDB.
- Cloudinary is used to make uploaded images available to the AI service.
- The repository includes architecture, subscription, API, setup, and limitation guides.

Representative service orchestration

```
POST /api/ai/image
  validate upload
  -> send image to Cloudinary
  -> call Groq vision model with hosted URL
  -> normalize model response
  -> return description to Angular

POST /api/subscription
  create Cashfree Sandbox request
  -> return authorization response
  -> update demo access state
```

The sequence summarizes the documented integration path rather than reproducing a source file.

Engineering review and production path

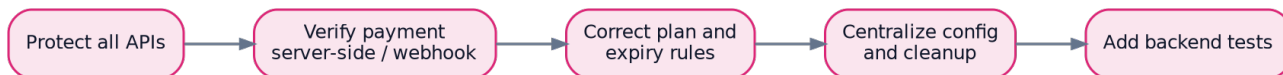
Engineering strengths

- Combines AI, authentication, media, persistence, and payment concepts.
- Handles three input modalities in one coherent product flow.
- Clear frontend/backend and external-service boundaries.
- Documentation openly distinguishes demo behavior from production requirements.

Current limitations

- Most AI and mutation routes need stronger JWT authorization.
- Subscription activation is trusted from the browser without webhook verification.
- Plan intervals and expiry handling contain documented inconsistencies.
- Backend automated tests and robust temporary-file cleanup are missing.

Recommended next steps



Why this is relevant to AI training work

- Demonstrates evaluation of model outputs across text, audio, and images.
- Requires prompt, format, and error normalization across different AI services.
- Shows how generated AI features must be constrained by auth, payment, and data-flow correctness.
- Identifies implementation defects instead of treating a successful demo as production quality.

Repository:

<https://github.com/SidheshwarSarangal/cashfree-ai-new-final>

Scope and provenance:

Prepared from repository architecture, user-flow, subscription, API/data, setup, and current-implementation documentation. Payment behavior is explicitly described as Cashfree Sandbox.

